ECM3412

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

COMPUTER SCIENCE

Examination, May 2019

Nature-Inspired Computation

Module Leader: Dr Ke Li

Duration: TWO HOURS

Answer question 1, and two out of the other three questions.
Question 1 is worth 40 marks. Other questions are worth 30 marks each.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

No electronic calculators of any sort are to be used during the course of this examination.

This is a CLOSED BOOK examination.

<u>SECTION A</u> (Compulsory Question)

<u>Question 1</u>

(a) Collectives in nature naturally exhibit a behaviour known as *emergence*.

(i) Briefly explain what the term emergence refers to in this context.

(2 marks)

(ii) Briefly explain why this behaviour is important for swarm intelligence algorithms such as PSO.

(2 marks)

(b) Evolutionary algorithms make use of selection operators to select members of the population to be parents of solutions in the next generation.

(i) Describe the process of *tournament selection* in an evolutionary algorithm.

(2 marks)

(ii) In tournament selection describe what happens to the selection pressure as the value of the tournament size *t* is increased and explain why this is the case.

(4 marks)

(iii) Explain why tournament selection is often preferred to other selection methods such as *roulette wheel selection*.

(4 marks)

(c) Explain how *local search* techniques differ from simple *hill-climbing* and why they may be better suited to solving problems with multi-modal fitness landscapes.

(6 marks)

(d) Table 1 gives the inputs $\mathbf{x}$ and output y values of an unknown mapping. Figure 1 gives a neural network structure that is able to realise such mapping.

- What is the name of this neural network?
- What are the weight settings for w_1 and w_2?
- What is the format of the activation function?

(5 marks)

Table 1: The input and output table

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

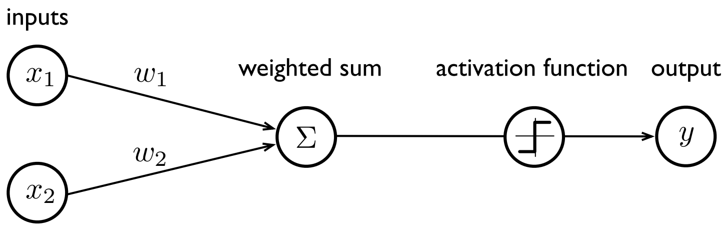

Figure 1: Example of neural network design.

(e) *Overfitting* is a major problem for predictive analytics and especially for neural networks. There can be various reasons that lead to the *overfitting*, e.g. the models constructed by the neural networks are quite complicated.

(i) Use a diagram to explain the relation between the *model complexity* and the *training error*, e.g. by using a trajectory.

(ii) In the same diagram, using a trajectory to explain the relation between the *model complexity* and the *generalisation error*.

(iii) In the same diagram, label the region that is regarded as *under-fitting*, and the region that is believed to be *overfiting*.

(10 marks)

(f) The *Game of Life* is a cellular automation devised by the British mathematician John Horton Conway in 1970. In particular, there are three different cell states.

(i) What situation will lead to the *death* of a cell?

(ii) In what situation, a dead cell will become alive, as known as *birth*?

(iii) In what situation, the cell does not change its state?

(5 marks)

(Total 40 marks)

SECTION B (Optional Questions)

Question 2

(a) Describe the main algorithm and, briefly, the experimental findings in the paper Wloch, K. and Bentley, P. J. (2004) “Optimising the Performance of a Formula One Car using a Genetic Algorithm” in the *Proceedings of PPSN Parallel Problem Solving from Nature-VIII, Lecture Notes in Computer Science, Volume 3242, 2004*.

(15 marks)

(b) Each nature-inspired algorithm has a different method of varying the extent to which it exploits or explores. For each algorithm below, describe the parameters that are modified to achieve different *exploitation* and *exploration* capability and explain how varying these alters the behaviour of the algorithm.

(i) Genetic Algorithms

(5 marks)

(ii) Particle Swarm Optimisation

(5 marks)

(iii) Ant Colony Optimisation

(5 marks)

(Total 30 marks)

Question 3

(a) Given a *multi-objective optimisation problem* defined as:

$$\begin{array}{ll} \text{minimize} & \mathbf{F}(\mathbf{x}) = (f_1(\mathbf{x}), \cdots, f_m(\mathbf{x}))^T \\ \text{subject to} & \mathbf{x} \in \Omega \end{array}$$

and two solutions $\mathbf{x^1}$ and $\mathbf{x^2}$, how can we said that $\mathbf{x^1}$ Pareto dominates $\mathbf{x^2}$?

(2 marks)

(b) Consider a *multi-objective optimisation problem* defined as:

$$\begin{array}{ll} \text{minimize} & \mathbf{F}(\mathbf{x}) = (f_1(\mathbf{x}), \cdots, f_m(\mathbf{x}))^T \\ \text{subject to} & \mathbf{x} \in \Omega \end{array}$$

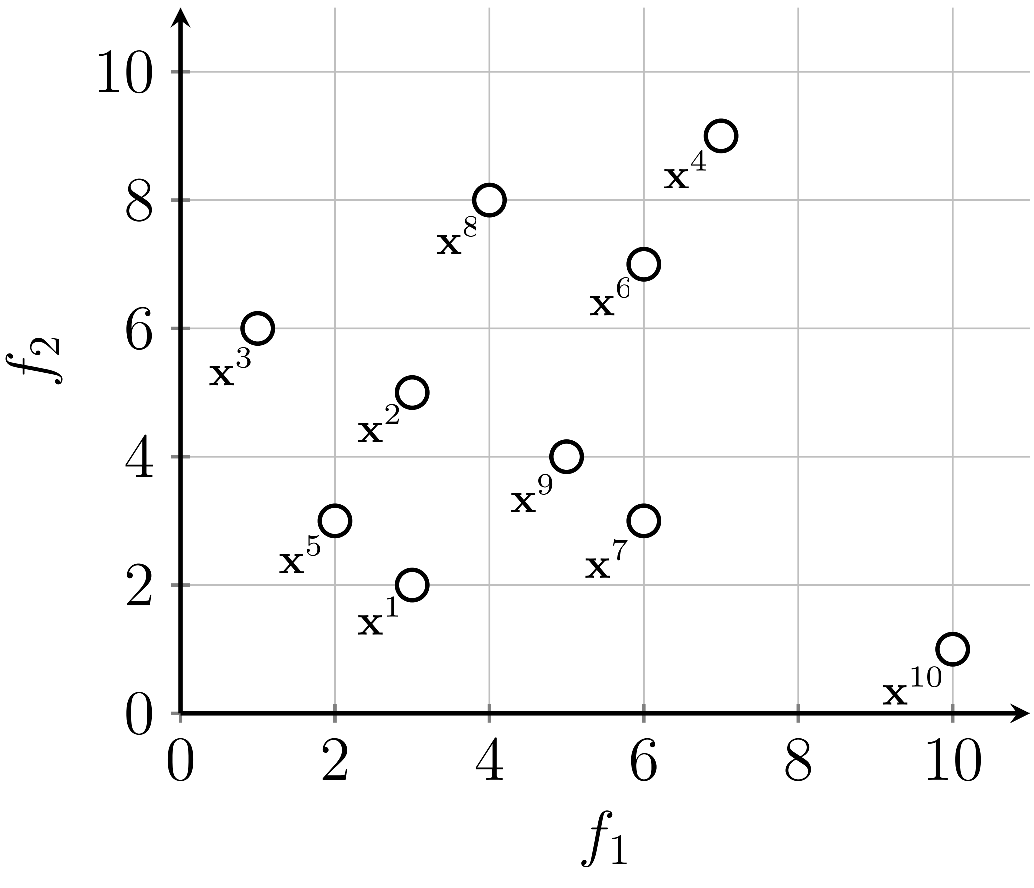

Figure 2: Population distribution in the objective space.

Figure 2 shows a set of solutions $S = \{\mathbf{x}^i\}_{i=1}^{10}$ distributed in the *objective space*. In particular, there are 10 solutions in total. The coordinates are

$\mathbf{F}(\mathbf{x}^1) = (3, 2)$, $\mathbf{F}(\mathbf{x}^2) = (3, 5)$, $\mathbf{F}(\mathbf{x}^3) = (1, 6)$, $\mathbf{F}(\mathbf{x}^4) = (7, 9)$, $\mathbf{F}(\mathbf{x}^5) = (2, 3)$, $\mathbf{F}(\mathbf{x}^6) = (6, 7)$, $\mathbf{F}(\mathbf{x}^7) = (6, 3)$, $\mathbf{F}(\mathbf{x}^8) = (4, 8)$, $\mathbf{F}(\mathbf{x}^9) = (5, 4)$, $\mathbf{F}(\mathbf{x}^{10}) = (10, 1)$.

(i) List the *non-dominated solutions* in S.

(4 marks)

(ii) In terms of convergence (measured by the Pareto dominance relationship), find the worst solution(s) in S.

(2 marks)

(iii) Consider using NSGA-II to evolve the population, non-dominated sorting is the first step in NSGA-II's environmental selection. How many non-dominated fronts can be obtained with respect to S?

(2 marks)

(iv) List the corresponding solutions that belong to each non-dominated front.

(4 marks)

(c) *Convergence* and *diversity* are the cornerstones of multi-objective optimisation. Most, if not all, evolutionary multi-objective optimisation algorithm designs consider achieving a balance between them, because we want to not only find solutions close enough to the Pareto front (i.e. convergence), but also want to make those solutions have a satisfactory distribution along the Pareto front (i.e. diversity). In actual optimisation, the underlying problems can be complex and lead to various challenges to the search process.

(i) What kind of properties or problem characteristics can make algorithms struggle to converge to the Pareto front/set? List *three* possible properties.

(3 marks)

(ii) What kind of properties or problem characteristics can make algorithms struggle to obtain a well distribution along the Pareto front/set? List *three* possible properties.

(3 marks)

(d) The *curse of dimensionality* is always the Achilles' heel of optimisation.

(i) Explain the possible reasons that lead to the curse of dimensionality in multi-objective optimisation from both decision space and objective space.

(2 marks)

(ii) List two challenges brought by the curse of dimensionality in the *decision space*.

(4 marks)

(iii) List two challenges brought by the curse of dimensionality in the *objective space*.

(4 marks)

(Total 30 marks)

Question 4

(a) Traditionally, perceptrons learn through the use of a learning algorithm (such as the delta rule), however, evolutionary algorithms (EAs) can also be used to optimise the weights within a perceptron.

(i) Describe how you would use an evolutionary algorithm to optimise the weights in a perceptron, paying particular attention to the representation, selection process, variation operators, fitness function, parameter settings and experimental design used to test your approach.

(10 marks)

(ii) The EA could also be used to optimise the weights of a multi-layer perceptron (MLP). Explain how you would change the EA representation to optimise the weights of an MLP and discuss any differences you would expect to see in algorithm behaviour as a result.

(5 marks)

(b) Ant colony optimisation (ACO) algorithm is an effective meta-heuristic for solving combinatorial optimisation problems like travelling sales man problem (TSP). The mathematical model of ACO is given as follows.

An ant is a simple computational agent in the ant colony optimisation algorithm, it iteratively constructs a solution for the problem, the k-th ant moves from city i to city j at the t-th iteration with probability $p_{ij}^k(t)$, and the equation of the transition rule is as follows.

$$\Delta_{ij}^k(t) = \begin{cases} \frac{[\tau_{ij}]^\alpha \cdot [\eta_{ij}]^\beta}{\sum_{h \in J_i^k} [\tau_{ih}]^\alpha \cdot [\eta_{ih}]^\beta} & : \text{if } j \text{ has not been visited,} \\ 0 & : \text{otherwise} \end{cases} \tag{1}$$

τ_{ij} is the amount of pheromone deposited for transition from city i to city j, and η_{ij} stands for heuristic information, which is typically the reciprocal of the distance. After the completion of a tour, each ant k lays a quantity of pheromone $\Delta_{ij}^k(t)$ on each edge (i, j) that it has used where Q is a constant, and L_k is the length of the tour constructed by ant k.

$$\Delta_{ij}^k(t) = \begin{cases} Q/L_k & : \text{if ant } k \text{ used edge}(i, j) \text{ in its tour,} \\ 0 & : \text{otherwise} \end{cases}$$

Also the pheromone τ_{ij}, associated with the edge joining cities i and j, is

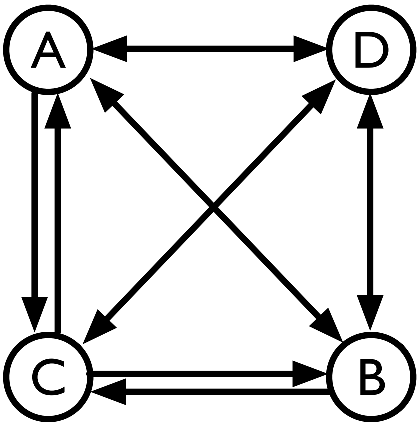

Figure 3: Four cities and their connections.

updated as where ρ is the evaporation rate, S is the number of ants.

$$\tau_{ij}(t+1) = (1-\rho) \cdot \tau_{ij}(t) + \sum_{k=1}^{S} \Delta_{ij}^{k}(t)$$

Considering a TSP problem as shown in Figure 3, there are four cities A, B, C, D. The distances between different cities are shown in the $DIST$ *matrix* as follows. Assuming the pheromone weight parameter $\alpha = 1$, the weighting parameter of the Heuristic factor $\beta = 2$, the constant $Q = 1$, the evaporation rate $\rho = 0.5$, Heuristic factor η is the reciprocal of the distance, initialisation pheromone $\tau_{ij}(t)$ is set to $\frac{1}{12}$.

$$DIST = (d_{ij}) = \begin{bmatrix} 0 & 1 & 0.5 & 1 \\ 1 & 0 & 1 & 1 \\ 1.5 & 5 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

(i) Calculate the pheromone matrix and heuristic factor matrix at the initialisation step.

(2 marks)

(ii) Calculate the probability of ants from the city A to the cities B, C, and D.

(6 marks)

(iii) Assume that $k = 4$, and the paths that the ants has searched in the first iteration is as follows.

- The path searched by the first ant: $A \to B \to C \to D \to A$.
- The path searched by the second ant: $A \to C \to D \to B \to A$.
- The path searched by the third ant: $A \to D \to C \to B \to A$.
- The path searched by the fourth ant: $A \to B \to D \to C \to A$.

The updated pheromone matrix is as follows:

$$\begin{bmatrix} 0 & 0.5139 & 0.3274 & 0.1667 \\ \tau_{BA}(t+1) & 0 & 0.2917 & 0.2639 \\ 0.2639 & 0.1667 & 0 & \tau_{CD}(t+1) \\ 0.2917 & \tau_{DC}(t+1) & 0.3889 & 0 \end{bmatrix}$$

Calculate $\tau_{BA}(t+1)$, $\tau_{CD}(t+1)$ and $\tau_{DC}(t+1)$ in the above pheromone matrix.

(7 marks)

(Total 30 marks)